Project Report
Risk Analytics Module

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Sneha - Documentation

1. Introduction

The Risk Analytics module is a critical component of the end-to-end research-to-production trading platform. Its purpose is to provide **daily risk assessment**, quantify downside exposures, and enable informed decision-making across strategies, portfolios, and execution workflows. Without robust risk measurement, trading strategies may appear profitable in backtests but fail under real-world stress.

This report outlines the scope, methodology, implementation, challenges, outputs, and contributions for the Risk Analytics module.

2. Scope and Objectives

The scope of the Risk Analytics module was aligned with the overall system goals:

- Deliver **daily risk metrics** (VaR, Expected Shortfall, Drawdowns, Sharpe Ratio).
- Provide **stress testing capabilities** for rate shocks, volatility spikes, and sector-specific drawdowns.
- Generate **exposure summaries** across sectors and asset classes.
- Produce **audit-ready outputs** including JSON, CSV, and plots for reproducibility.
- Ensure seamless **integration with backtesting, portfolio optimization, and dashboard modules**.

3. Methodology

3.1 Risk Metrics

We implemented standard industry-grade risk measures:

- **Value at Risk (VaR):**

  $$\text{VaR}_\alpha = -\text{Quantile}(1-\alpha)(R_t)$$

  Measures the worst expected loss at confidence level (\alpha).

- **Expected Shortfall (ES):**

  $$DD_t = \max(NAV_1 \ldots t) NAV_t - \max(NAV_1 \ldots t)$$

  Captures the **average tail loss**, improving upon VaR.

- **Drawdowns:**

  $$SR_t = \sigma(R_t) E[R_t - R_f] \times 252$$

  Monitors cumulative downside risk relative to portfolio peaks.

- **Sharpe Ratio (Rolling):**

  [
  SR_t = \frac{E[R_t - R_f]}{\sigma(R_t)} \times \sqrt{252}
  ]

  Evaluates risk-adjusted returns against the risk-free rate.

Metric	Example Value
VaR 95% Daily	0.00356
ES 95% Daily	0.00296
Sharpe Ratio	1.25
CAGR	8.3%
Maximum Drawdown	-6.65e-06

Table 1: Sample Risk and Performance Metrics

3.2 Stress Testing

To evaluate resilience under shocks:

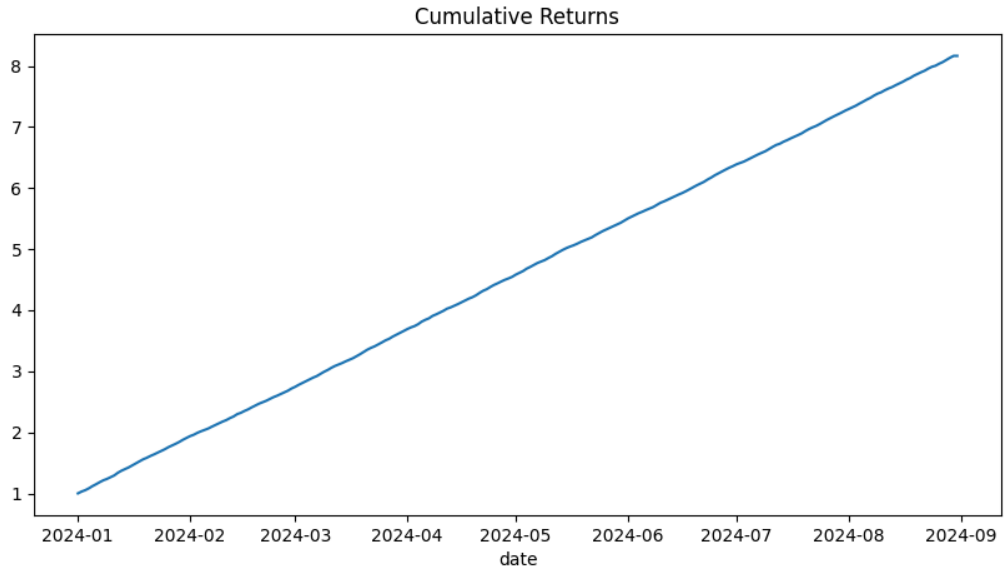
- **Sector Shock:** Reduces valuation of a specific sector (e.g., -15% Technology exposure).
- **Rates Shock:** Uses credit-rating-based duration mapping to estimate interest rate sensitivity.
- **Volatility Shock:** Scales portfolio volatility (e.g., 1.5× regime shift).

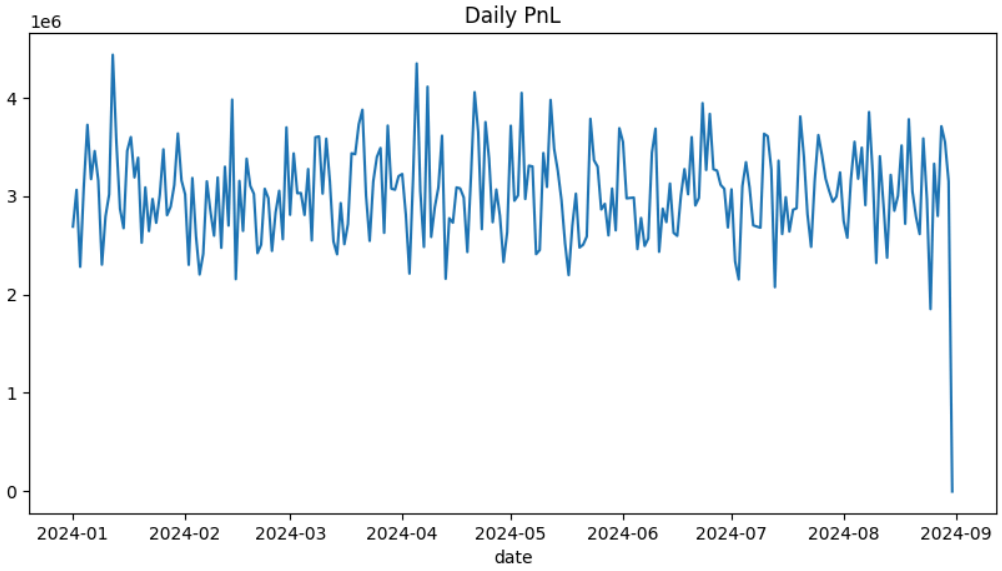
Scenario	Impact %	Impact USD
Sector Shock	-2.5%	-$2,500,000
Interest Rate Shock	-1.8%	-$1,800,000
Volatility Spike	-3.2%	-$3,200,000

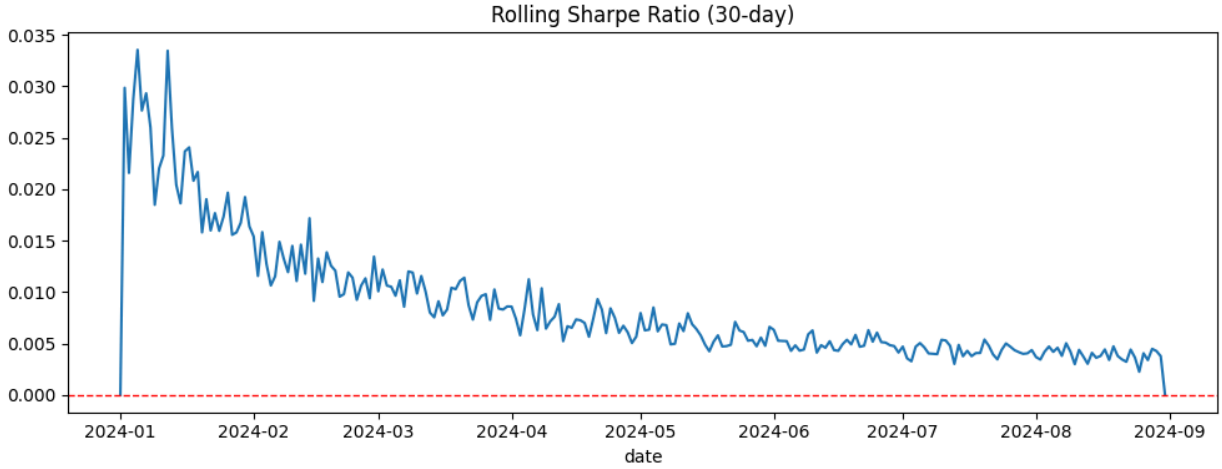
Table 2: Example Stress Test Results

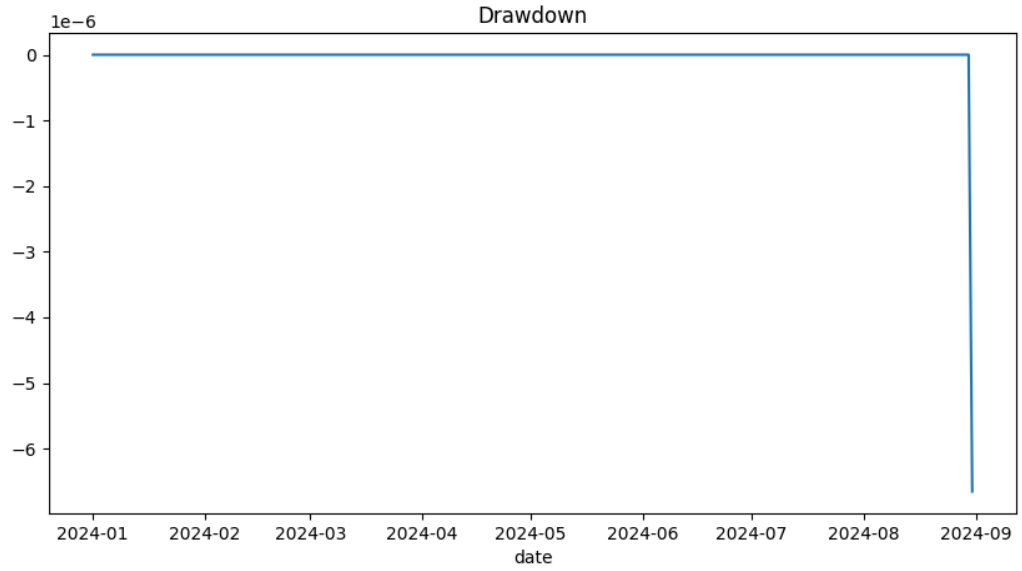
3.3 Implementation Path

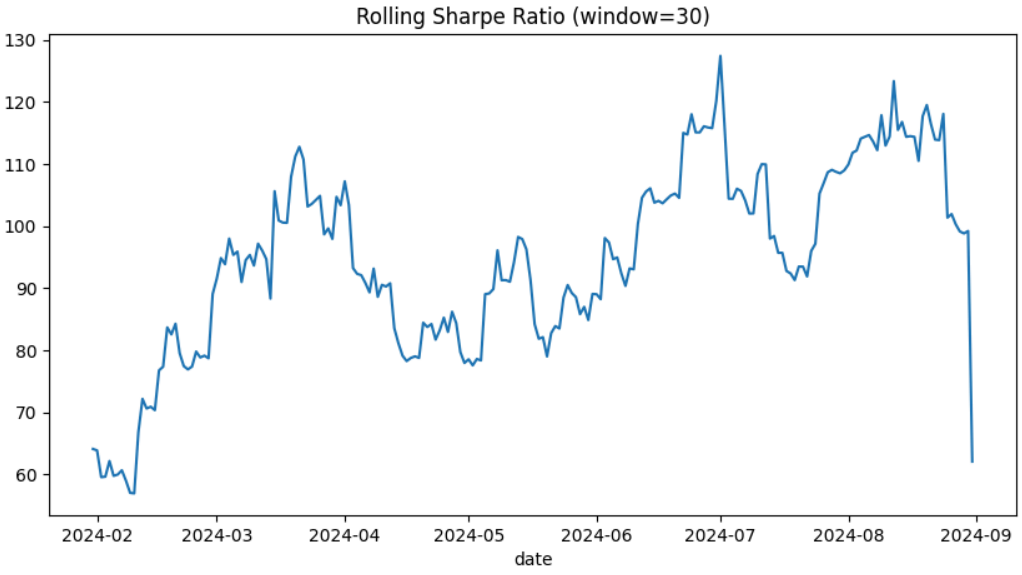
- Built on **Python with Pandas, NumPy, Matplotlib**.
- Modular design with dedicated scripts for **metrics, stress testing, reporting, and orchestration**.
- Optimized for **scalability** (30,000 trades × 1,500 instruments).
- Ensured **audit readiness** via Git commit hashes, config snapshots, and data checksums.




Daily PnL
1e6
4
3
2
1
0
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07
2024-08
2024-09
date


Rolling Sharpe Ratio (30-day)
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07
2024-08
2024-09
date





4. Key Outputs

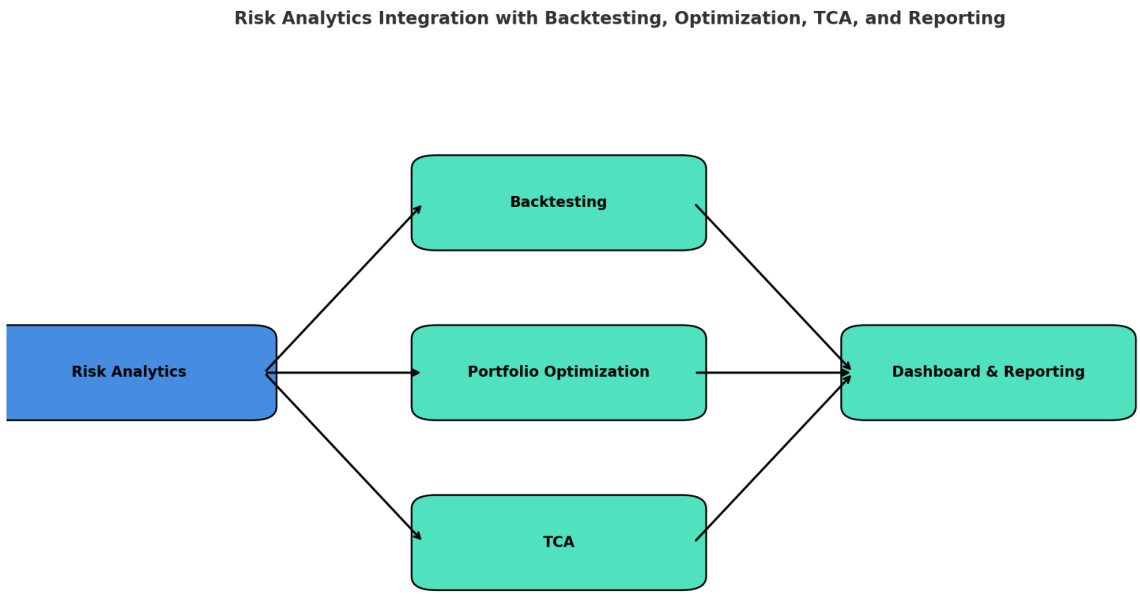
The pipeline generates the following outputs daily:

- **CSV Files**: Daily PnL, returns, strategy-level returns, drawdowns.
- **JSON Files**: VaR/ES metrics, performance summaries, exposures, stress test results, audit logs.

- **Plots**: Daily PnL trends, cumulative NAV, rolling Sharpe ratio, drawdown curves.
- **Audit Trail**: Complete metadata linking data, configs, and code versions.

These outputs are consumed by:

- **Backtesting team** – to validate strategies under realistic risk limits.
- **Portfolio Optimization team** – to apply exposure, volatility, and drawdown constraints.
- **Dashboard team** – to display live performance and risk monitoring.



5. Challenges Faced and Solutions

1. **Data Schema Mismatch**
    - Problem: Trading and instrument datasets lacked duration and volatility fields.
    - Solution: Introduced **flexible mapping and defaults** (e.g., duration from credit rating).

2. **Volatility Normalization**
    - Problem: Inconsistent scaling across instruments.
    - Solution: Standardized all volatilities into decimal form (0.02 = 2%).
3. **Stress Test Realism**
    - Problem: Naïve PnL-based shocks were unrealistic.
    - Solution: Switched to **market value–based shocks** for institutional accuracy.
4. **Integration with Other Teams**
    - Problem: Ensuring outputs were compatible downstream.
    - Solution: Standardized formats (JSON/CSV) and aligned schemas with optimization & backtesting teams.

6. Contributions

6.1 Sushmitha Balakumar - Risk Analytics Lead (Design & Implementation)

End-to-End Ownership: Designed, developed, and implemented the **Risk Analytics module** of the trading research-to-production system.

Technical Implementation:

- Built the **core pipeline** (main.py, metrics.py, stress.py, reporting.py).
- Implemented **daily risk metric calculations** (VaR, Expected Shortfall, Drawdowns, Sharpe Ratio).
- Developed **stress testing scenarios** (sector shocks, rate shocks, volatility spikes).
- Designed **exposure analysis** (sector, asset class) for downstream risk-aware decision-making.
- Ensured **audit-readiness** with versioning, data snapshots, and reproducible outputs.

Optimization & Problem Solving: Addressed dataset gaps (missing duration/volatility), normalized inputs, and improved realism in stress modeling.

Integration: Delivered standardized outputs (CSV, JSON, plots) that feed directly into **backtesting, portfolio optimization, and dashboards**.

Industry Alignment: Designed the module with **institutional-grade practices** in mind (scalability, reproducibility, risk governance).

6.2 Sneha's Contributions - Documentation

- Structured and prepared the **written documentation** for the Risk Analytics module.
- Summarized methodologies, formulas, and workflows in a clear and professional manner.
- Ensured the report aligns with academic and professional standards for submission.

7. Conclusion

The Risk Analytics module successfully delivers **daily risk insights, stress test evaluations, and audit-ready reports**, serving as the backbone for monitoring and controlling portfolio risks. My work ensured that the pipeline is **production-grade, scalable, and integrable** with other modules, while Sneha provided documentation support.

This module not only meets the **scope and functional requirements** of the project but also ensures that downstream teams (backtesting, optimization, dashboard) can rely on accurate, standardized, and reproducible risk data.